

Semantic Drill-Down — RAG Pipeline

How the knowledge-base retrieval works, end to end

Current implementation overview · grounded, cited answers from the organization's own manuals

1. What this is

RAG = **Retrieval-Augmented Generation**. Instead of letting an AI answer from its general training, we first **retrieve** the relevant passages from the organization's own approved manual, then **augment** the AI's prompt with that verified text, so the **generated** answer is grounded in real documentation and can cite the exact source page.

In this product a technician points at equipment and clicks a part. The system identifies it, looks it up in the uploaded manual, and returns the answer **with a citation** (e.g. "IFB MACHINE.pdf, page 7") instead of a guess.

Analogy (non-technical): a fast librarian reads the manual once — every page, table, and diagram — and files it on index cards. Later it instantly pulls the cards that answer your question and tells you which page they came from.

2. Two phases

The pipeline has two clearly separated phases. Most quality issues live in one or the other.

Phase	When	What it does
Ingestion	Once per PDF (offline)	Read the manual, break it into searchable pieces, store them with two search 'fingerprints'.
Retrieval + Generation	Every drill (online)	Turn the user's target into a query, find the right page, cite it, and ground the generated answer.

3. Ingestion pipeline (building the index)

```
PDF upload
| kb_id = sha256(pdf)[:16]    (idempotent: re-upload rebuilds)
v
Docling -> layout-aware parse: headers / text / tables / figures (with page numbers)
|
+-- TEXT track    -> word-window chunks (320 words, 60 overlap), section-prefixed
+-- TABLE track  -> one chunk per ROW (header-prefixed) + table summary
|                  (table-of-contents tables are detected & skipped)
+-- FIGURE track  -> Qwen2.5-VL vision model writes a TEXT description of the diagram
v
Each chunk -> TWO vectors:
- dense (BAAI/bge-small-en-v1.5, 384-dim) = meaning
- sparse (BM25 via fastembed)             = exact keywords / part numbers
v
Qdrant (local) -> collection kb_{id}, named vectors {dense + sparse}
                  payload: text, page_id, page_num, source_name, zone
```

Why three tracks? A manual is not flat text. Tables and diagrams carry critical information, so each content type is handled its own way: prose is windowed, tables are split row-by-row so a single row is a precise answer, and diagrams are converted to searchable text by a local vision model.

4. Retrieval + generation pipeline (answering a drill)

```
User clicks a region (optionally types a label, e.g. 'Control Panel')
v
Vision model (Qwen2.5-VL) describes the clicked region
v
QUERY = label (typed) -> object name (1st sentence) -> analysis slice
                  (NOT the verbose generation prompt -- it invents detail not in the manual)
v
Embed query: dense (BGE) + sparse (BM25)
v
Qdrant hybrid search -> RRF fusion of the two ranked lists
v
Re-join all chunks on the winning page (by page_id) into one context block
```

```

      v
Confidence = best dense COSINE on the page; if < 0.60 -> 'No strong manual match'
      v
Inject verified page text into the generation prompt + show citation
      'IFB MACHINE.pdf, page 7 - 70% match'

```

Hybrid search runs meaning-search (dense) and keyword-search (BM25) in parallel and merges them with Reciprocal Rank Fusion (RRF), so exact terms like part numbers and fault codes are caught alongside semantic matches.

Gating: in Generic mode (no knowledge base selected) this whole block is skipped — no retrieval, no citation. KB retrieval runs only when a manual is selected.

5. Key design decisions & trade-offs

Decision	Why	Trade-off
Hybrid dense + BM25	Meaning + exact codes/part numbers	Two indexes to maintain
BGE-small (local)	CPU-friendly, 512-token context	Lower ceiling; compressed score range
RRF fusion	Merges rankings w/o normalizing scores	Rank score != true relevance
One chunk per table row	Precise row-level answers	More chunks
VLM caption for figures	Diagrams become searchable text	Caption can miss detail
Page re-join	Full context + clean page citation	Coarser than chunk-level cite
Label-first query	Concise term matches manual wording	Pure clicks less precise
Confidence floor (0.60)	Honest 'no match' > wrong citation	Can't fully separate weak vs junk
Local Qdrant (file mode)	Simple, private, on-prem	Single-process; not concurrent yet

6. Honest limitations

- **Query quality dominates.** Retrieval is only as good as the query; a typed component label is the most reliable.
- **Compressed similarity.** BGE cosine clusters ~0.6–0.9, so a weak real match can overlap off-topic noise; a re-ranker would fix this.
- **Coverage is a hard limit.** RAG can only cite what the manual contains (a mechanical manual has no 'LCD / circuit board').
- **Captioning, not true multimodal.** Diagrams are converted to text once at ingestion; fine print can be lost.
- **Single-process store.** Local file-mode Qdrant needs server mode to scale to many concurrent users.

7. Anticipated questions

Manager

How accurate is it? Cites the right page with high confidence when the manual covers the part; when unsure it says 'no strong match' instead of guessing.

What if it's wrong? Every citation shows a confidence %, and weak matches are suppressed. The user always sees and can verify the source.

Is our data safe? Everything runs locally/on-prem — vision model, embeddings, and vector DB. The manual never leaves the machine.

What does it cost? No per-query cloud cost. The heavy work happens once at upload; queries are cheap.

AI architect

Chunking? 320-word windows / 60 overlap for prose; row-level for tables; whole-caption for figures; section headers prefixed; sized to the embedder's 512-token budget.

Why BGE-small? Local/CPU/private constraint; pluggable — a larger or cloud embedder would improve separation.

Fusion? RRF over independent dense + sparse prefetches; confidence taken from dense cosine, not the fused rank.

Re-ranking? None yet — a cross-encoder re-ranker on top-k is the natural next precision upgrade.

Grounding/hallucination? Generation is grounded on retrieved verified text with a citation; below-threshold matches are suppressed.

Scaling? Move local Qdrant to server mode; batch/queue embedding + VLM for concurrency. Multi-manual org-wide search is an extension.

Evaluation? Known-answer queries with expected page + score; automated precision/recall regression is a gap to close.

Stack: Docling (parse) · Qwen2.5-VL (figure captioning, region analysis) · BAAI/bge-small-en-v1.5 (dense) · BM25/fastembed (sparse) · Qdrant (hybrid vector store). All local.